

Nate Woodward

LC T P Spring symposium: Theoretical Physics and AI

Making a SFT Dataset

- **Rejection Sample: 5 responses per problem in synthetic QFT train & val.,
only keep correct**

24,918 correct respns, ~2000 Mistaken verified Chain Thought (CoT)

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5

- **Select three candidate teacher models:**
oss-120b, Qwen3-30b, Qwen-3.5-122b



GPT-OSS

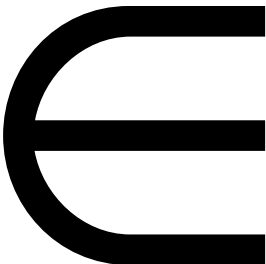
120B



Qwen3



Qwen3.5



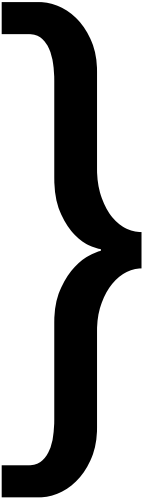








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Making a SFT Dataset

- Select **three candidate teacher models**:
oss-120b, Qwen3-30b, Qwen-3.5-122b



$$? \in \left\{ \text{GPT-OSS}_{120B}, \text{Qwen3}, \text{Qwen3.5} \right\}$$

- Rejection Sample: **5 responses per problem** in synthetic QFT train & val.,
only keep correct
- 24,918 correct responses, ~200M tokens of verified Chain of Thought (CoT)

Choosing the Teacher

Train base model on QFT Easy CoT from each teacher + all combined

Student Model	Base	SFT Teacher Model (CoT Source)			
		OSS-120B	Qwen3-30B	Qwen3.5-122B	Combined
DeepSeek-7B	40.2	50.4	59.7	47.3	50.4